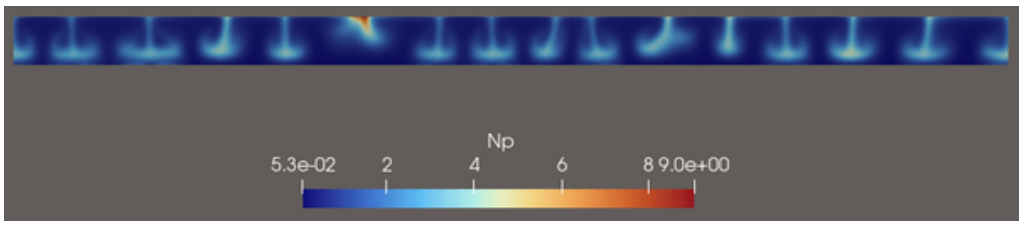
Study of Microalgae Bioconvection

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Based on the PhD thesis of Anaëlle GIVAUDAN

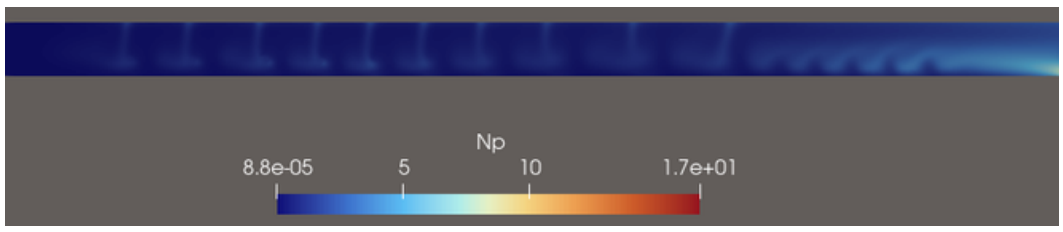
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Introduction:

Bioconvection is a self-organisation phenomenon which here resembles a Rayleigh–Bénard-type instability. It arises from the accumulation of microalgae at the surface due to gyrotaxis, creating a dense, gravitationally unstable layer. Once the critical Rayleigh number is exceeded, the system collapses under gravity, forming plumes that are continuously regenerated.



Without light

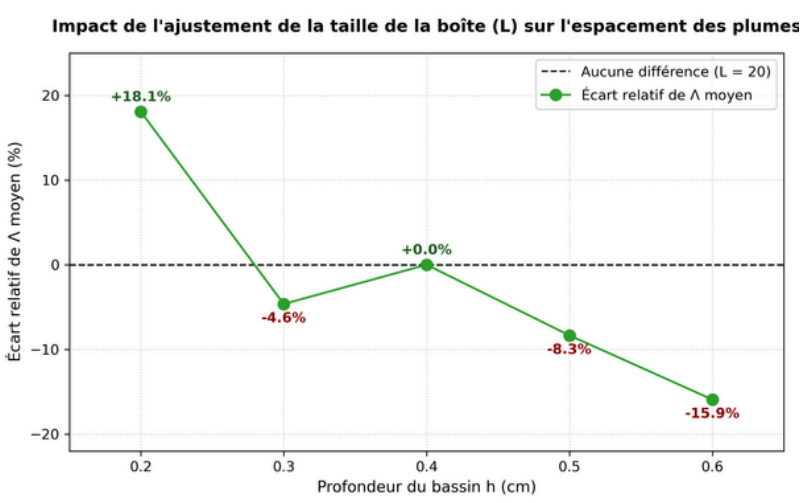


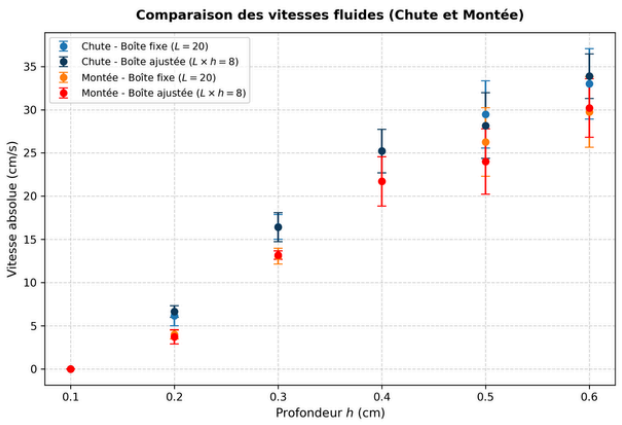
With light

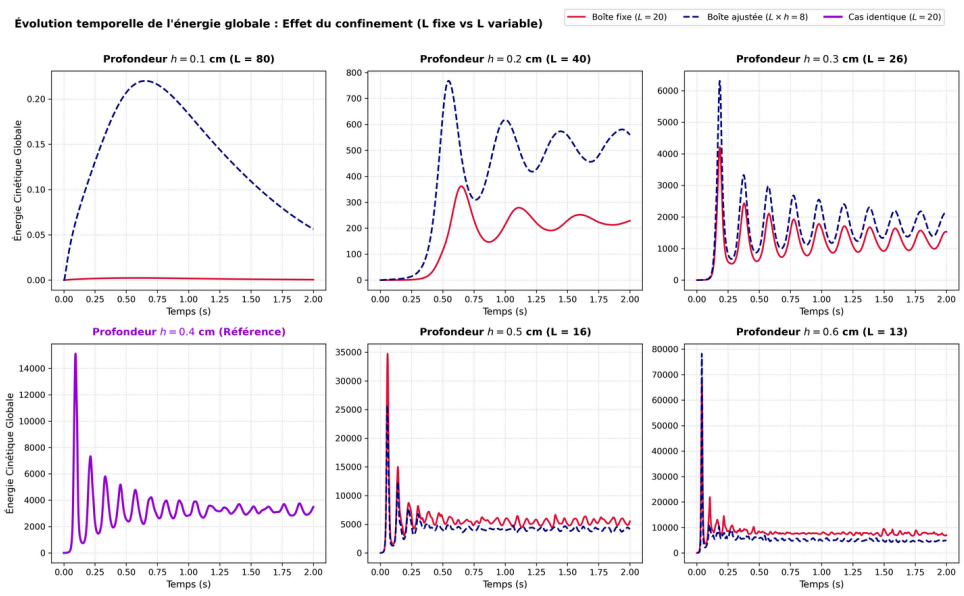
Objectives:

- Assess the influence of a variable viscosity μ, depending on the local algae concentration, compared with a constant-viscosity model (without light)
- Influence of the domain geometry (edge effects) and comparison between fixed and variable L (without light)
- Influence of phototaxis on the plume migration velocity (with light)

Edge effects and fixed vs. variable L:



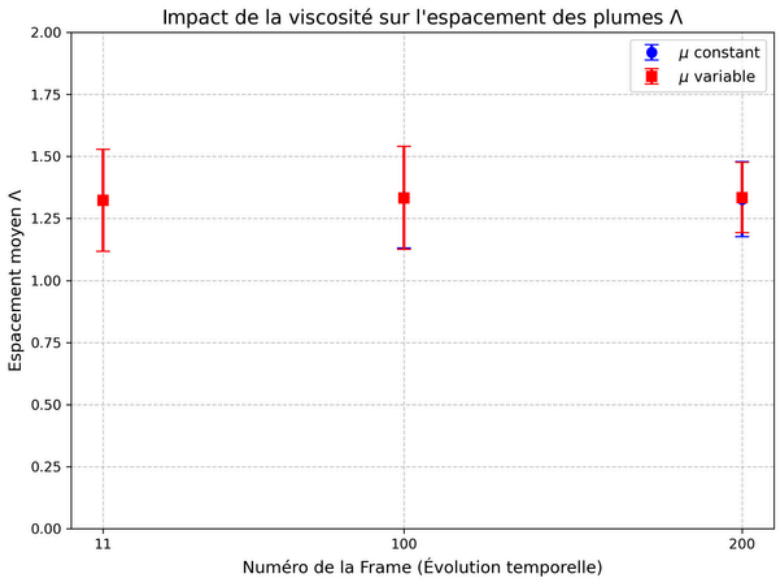




The experimental geometry imposes: L × h = 8

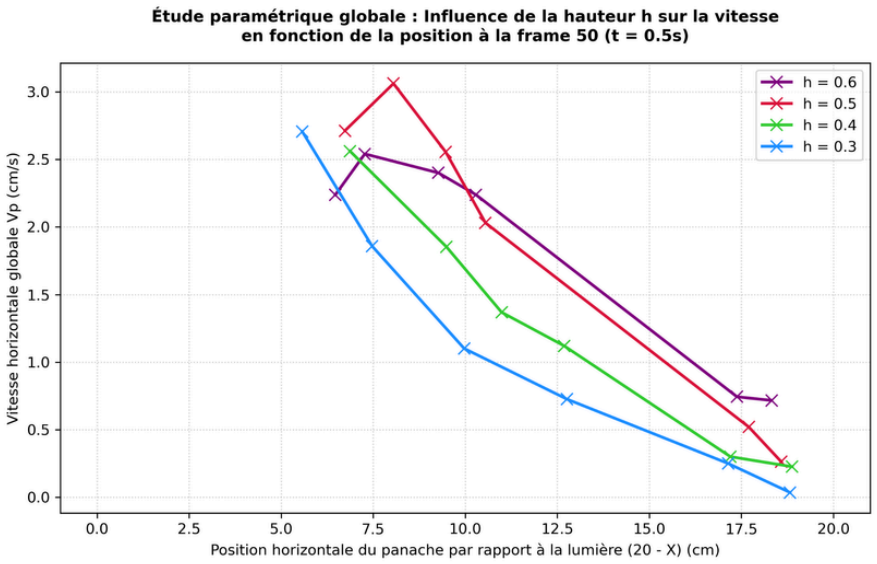
- Fixing L = 20 introduces a geometric bias at the extremes (confinement imposed on the convection rolls); adjusting L is therefore essential a priori
- The bioconvection dynamics (sinking and rising velocities, vorticity intensity) remain governed by h and the fluid properties
- Energy monitoring confirms that the critical Rayleigh threshold is crossed between h = 0.1 and h = 0.2 (onset of plumes)

Constant vs. variable μ :



The constant-viscosity approximation is relevant: it captures the essential physics of the problem while reducing the computational cost, without loss of accuracy.

Migration velocity:



The migration velocity increases with the basin depth h and with proximity to the light source: convection and phototaxis reinforce each other.

Conclusion:

This bioconvection results from a coupling between the physical parameters of the system, such as the basin depth h or the boundary conditions, and biological and environmental factors such as the algae swimming speed and light.